

TARPITA KARNAM

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EDUCATION

Vellore Institute of Technology

B.Tech in Computer Science (Health Informatics) — CGPA: 8.50

2023 – 2027

Bhopal, India

Loyola School

12th Grade — 87%

2023

Jamshedpur, India

Loyola School

10th Grade — 90.2%

2021

Jamshedpur, India

TECHNICAL SKILLS

Languages: C++, JavaScript, SQL

Frontend: HTML5, CSS3, React.js, TailwindCSS

Backend: Node.js, Express.js, REST APIs, WebSockets

Databases & Cloud: MongoDB, Firebase

Tools & Platforms: Git, GitHub, VS Code, Vite, Canva

Core Concepts: OOPS, DBMS, Role-Based Access Control (RBAC), JWT Authentication

EXPERIENCE

Frontend Developer Intern

Dec 2025 – Jan 2026

Info Bharat Interns

Remote

- Developed **Teccho Hotel Booking Web App** (teccho-hotel-booking-app.vercel.app), a complete hotel booking platform with responsive UI and seamless booking workflow.
- Implemented room listings, booking and checkout flow, authentication system, and order history management.
- Built scalable state management using **Redux Toolkit** and structured routing with **React Router**.
- Designed responsive layouts using **Tailwind CSS** to ensure cross-device compatibility.
- Improved frontend architecture through reusable components and optimized state patterns.

PROJECTS

EduGuide – College Discovery & Predictor

2026

- **Technologies:** React 19, TypeScript, Vite
- Engineered an educational single-page application (2,100+ LOC) featuring 8 interactive screens to streamline college discovery, comparison, and programmatic student workflows.
- Developed an intelligent admission predictor analyzing 60+ historical seat cutoff configurations across 5 admission categories and 3 exam pathways to calculate and tier admission probabilities.
- Architected a normalized database for 11 institutions and 31 academic programs, optimizing production build size to an 85.8 kB gzipped bundle through strategic module transformation and memoization.

HealConnect – IoT & AI Healthcare Platform

Jan 2026 – Present

- **Technologies:** Next.js, Firebase, Firestore, WebSockets, REST APIs, Gemini AI, C++
- Developed core C++ firmware for wearable IoT hardware, establishing a low-latency data pipeline to stream live telemetry via WebSockets and Firestore listeners.
- Architected a comprehensive full-stack platform integrating Role-Based Access Control (RBAC) to securely isolate data across Patient, Doctor, and Admin dashboards.
- Engineered 3 secure serverless REST APIs and integrated external LLMs (Gemini) via asynchronous polling loops to automate clinical data processing.

CERTIFICATIONS

Microsoft Certified: Azure Data Fundamentals (DP-900)

EXTRA-CURRICULAR & LEADERSHIP EXPERIENCE

Linpack Club

Sep 2024 – March 2026

Social Media Lead

VIT Bhopal

- Managed the club's digital presence and promotional strategies, substantially increasing online engagement and community reach.
- Coordinated with core technical teams to advertise events, maximizing student outreach and participation.

Notion VIT Bhopal

Feb 2024 – Dec 2025

Social Media Team Co-Lead

VIT Bhopal

- Directed social media campaigns for technical workshops, successfully boosting event visibility and attendee registrations.
- Streamlined digital communication and marketing workflows to maintain consistent community engagement.